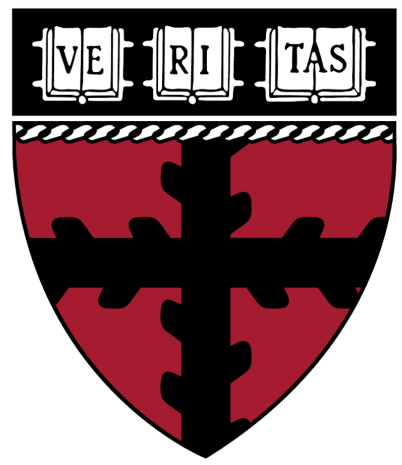




When the Manifold Bends, the Model Lies? Geometric Predictors of Hallucinations in LLMs



Mohamed Zidan Cassim, Sein Yun, Christopher Perez

We find hallucinations concentrate in distinct geometric regions of embedding space, making centrality a universal predictor of risk across 10 frontier models ($p < 0.001$), with curvature adding additional but less universal signal.

Motivation & Problem

- LLMs hallucinate confidently, even in high-stakes domains like medicine, law, and security.
- Current methods (like RAG or Self-Consistency) try to fix errors after generation, which is computationally expensive and model-specific.
- Missing: a model-agnostic, pre-generation warning signal.

Key Question

Can the geometry of prompt embeddings predict hallucination risk before the model generates an answer?

Methodology

- Dataset & Pipeline:**
 - Dataset:** 449 prompts across 5 categories (Factual, Nonexistent, Impossible, Ambiguous, Borderlines).
 - Models:** 10 frontier LLMs (GPT-5/4, Claude Opus/Sonnet/Haiku, Llama, Mixtral, Qwen)
 - PROMPTS (449) → RESPONSES (10 LLMs) → GEOMETRY + CONSENSUS JUDGING
- Geometric Features:** Extracted from text-embedding-3-small space
 - Centrality:** Distance to global centroid
 - Curvature:** Local manifold flatness, PCA residual
 - LID & Density:** Local complexity and sparsity
- Ground Truth (Consensus):**
 - Panel:** GPT-5 + Claude Opus + Llama 4
 - Method:** Majority vote (2/3 agreement)
 - Reliability:** 96% confidence, 90% human agreement

Impact

Using geometry alone, we flag the **top 30% most risky prompts** → prevent **~89% of hallucinations**. Works **model-agnostically** and **pre-generation**.

Results

